

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular & Supplementary Winter Examinations – 2025 Course: M. Tech. Branch: Computer Science and Engg & Allied Semester :III Subject Code & Name: MTCSSMD302B - Ethical Hacking Max Marks: 60 Date: 30/01/2026 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Define Ethical Hacking. Explain its objectives and phases.	CO1 BT2	6
B)	Explain the TCP/IP protocol stack with a neat labeled diagram.	CO2 BT2	6
C)	Explain IPv4 addressing and subnetting with a suitable numerical example.	CO3 BT2	6
Q.2	Solve Any Two of the following.		12
A)	Explain the procedure for installation of attacker and victim systems in a controlled lab environment.	CO3 BT2	6
B)	Describe information gathering techniques using: WHOIS, dig, and Netcraft.	CO3 BT3	6
C)	Explain how NMAP is used for host discovery and port scanning.	CO3 BT3	6
Q. 3	Solve Any Two of the following.		12
A)	Explain vulnerability scanning using NMAP and Nessus. Compare both tools.	CO4 BT4	6
B)	Describe system hacking with reference to password cracking and privilege escalation.	CO3 BT3	6
C)	Explain ARP spoofing and MAC flooding attacks with neat diagram and preventive measures.	CO4 BT4	6
Q.4	Solve Any Two of the following.		12
A)	Compare private-key encryption and public-key encryption with suitable examples.	CO2 BT4	6
B)	Explain cryptographic hash function and digital certificates with	CO2 BT2	6

	applications.		
C)	A company faces repeated DNS spoofing and email phishing attacks. Explain the attack mechanism and propose suitable security solutions.	CO4 BT5	6
Q. 5	Solve Any Two of the following.		12
A)	Explain packet sniffing using Wireshark and Burp Suite. How can intercepted credentials be protected?	CO5 BT3	6
B)	Describe social engineering and Denial of Service attacks with real-world scenarios and defenses.	CO5 BT4	6
C)	A web application is vulnerable to SQL Injection and XSS. Explain how Metasploit can be used to exploit these vulnerabilities and suggest countermeasures.	CO5 BT5	6
	*** End ***		

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